Practice workbook · Velocity and displacement by integrating

Integrating acceleration to get velocity, and velocity to get displacement, using starting conditions for the constants.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-28

1. Question 5 · part c · 4 marks

2022 JAN · Paper 02

Worked-example retry · Attempt part c only.

5 A particle P is moving along the x-axis. At time t seconds ($t \geqslant 0$) the velocity, v m/s, of P is given by $v = 3t^2 - 23t + 30$

(a) Find the values of t when P is instantaneously at rest.

(3)

At time t seconds the acceleration of P is a m/s^2

(b) Find the range of values of t for which $a > 0$

(2)

When $t = 0$, P is at the point with coordinates $(d, 0)$

Given that, when $t = 8$, P is at the point with coordinates $(26, 0)$

(c) find the value of d

(4)

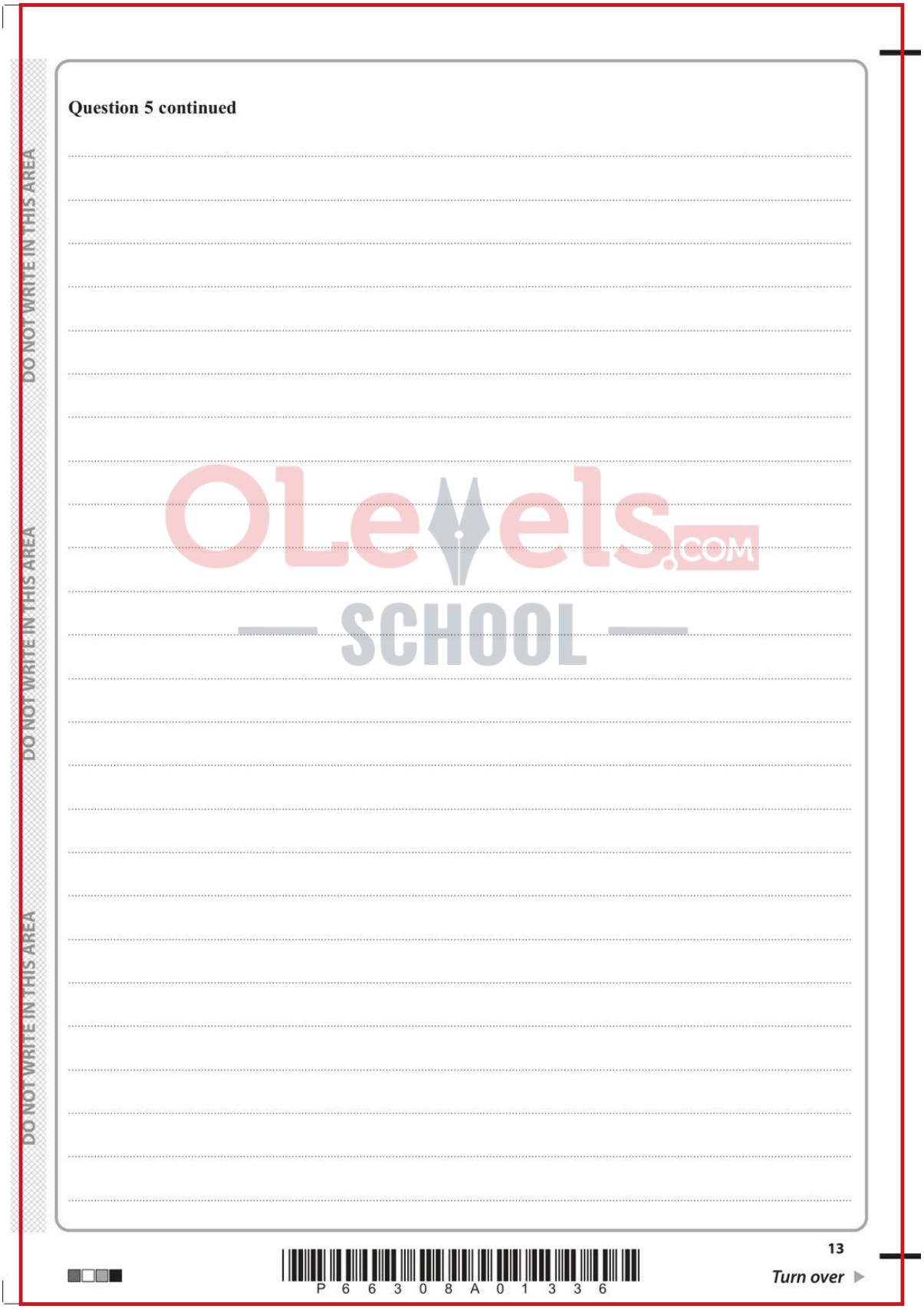
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1. Question 5 · part c · 4 marks

2022 JAN · Paper 02

Worked-example retry · Attempt part c only.



1. Question 5 · part c · 4 marks

2022 JAN · Paper 02

Worked-example retry · Attempt part c only.

Question 5 continued

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1. Question 5 · part c · 4 marks

2022 JAN · Paper 02

Worked-example retry · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 9 marks)

1. Question 5 · part c · 4 marks · Working and answer

Working and answer

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2. Question 3 · part a.ii · 4 marks

2023 NOV · Paper 02

Further practice · Attempt part a.ii only.

3 A particle P moves along the x-axis.

At time t seconds $(t \geqslant 0)$ the acceleration, a m/s^2, of P is given by $a = 6t - 16$

When $t = 0$, P is at the origin and is moving with velocity 12 m/s.

(a) Find an expression in terms of t for

(i) the velocity of P at time t seconds

(ii) the displacement of P at time t seconds.

(4)

(b) Hence find the time at which P first returns to the origin.

(3)

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P 7 3 5 8 6 A 0 6 3 2

2. Question 3 · part a.ii · 4 marks · Working and answer

Working and answer

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3. Question 4 · part a · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

4 A particle P moves along the x-axis. At time t seconds ($t \geqslant 0$) the acceleration, a m/s^2, of P is given by $a = 6t - 12$

When $t = 0$, P is at rest at the origin.

(a) Find the velocity of P when $t = 2$

(3)

At time T seconds, $T > 0$, P is instantaneously at rest.

(b) Find the value of T.

(2)

(c) Find the distance travelled by P in the first 8 seconds of its motion.

(3)

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3. Question 4 · part a · 3 marks · Working and answer

Working and answer

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4. Question 5 · part a · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part a only.

5 A particle P is moving along the x-axis.
At time t seconds $(t \geqslant 0)$ the acceleration, a m/s^2, of P is given by $a = 3t - 4$

When $t = 0$, P is at rest.

(a) Find the velocity of P when $t = 4$

(3)

At time T seconds, $T > 0$, P is instantaneously at rest.

(b) Find the value of T

(2)

When $t = 0$, P is at the point with coordinates $(-10, 0)$

(c) Find the displacement of P from the origin when $t = 3$

(4)

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4. Question 5 · part a · 3 marks · Working and answer

Working and answer

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5. Question 5 · part c · 4 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

5 A particle P is moving along the x-axis.
At time t seconds $(t \geqslant 0)$ the acceleration, a m/s^2, of P is given by $a = 3t - 4$

When $t = 0$, P is at rest.

(a) Find the velocity of P when $t = 4$

(3)

At time T seconds, $T > 0$, P is instantaneously at rest.

(b) Find the value of T

(2)

When $t = 0$, P is at the point with coordinates $(-10, 0)$

(c) Find the displacement of P from the origin when $t = 3$

(4)

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5. Question 5 · part c · 4 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

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Question 5 continued

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(Total for Question 5 is 9 marks)

5. Question 5 · part c · 4 marks · Working and answer

Working and answer

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6. Question 8 · part whole · 8 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

8 A particle P moves along the positive x-axis. At time t seconds ($t \geqslant 0$) the velocity, v m/s, of P is given by $v = 3 + 5t - 2t^2$

At time t seconds, P is at the point with coordinates $(x, 0)$.

Given that at time $t = 0$, P is at the point with coordinates $(5, 0)$, find the maximum value of x, justifying that this is a maximum value.

(8)

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6. Question 8 · part whole · 8 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 8 continued

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6. Question 8 · part whole · 8 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 8 continued

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P 6 0 4 7 5 A 0 2 2 3 6

6. Question 8 · part whole · 8 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 8 continued

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(Total for Question 8 is 8 marks)

6. Question 8 · part whole · 8 marks · Working and answer

Working and answer

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7. Question 2 · part b · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

2 A particle P is moving along a straight line, which passes through the fixed point O.

At time t seconds ($t \geqslant 0$), the velocity, v m/s, of P is given by

$$v = t^2 - 3t + 4$$

At time t seconds the acceleration of P is a m/s^2

(a) Find an expression for a in terms of t

(2)

The displacement of P from O is 7 m when $t = 2$

(b) Find the exact displacement of P from O when $t = 4$

(5)

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7. Question 2 · part b · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.



Question 2 continued

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(Total for Question 2 is 7 marks)

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7. Question 2 · part b · 5 marks · Working and answer

Working and answer

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8. Question 7 · part b · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

7 A particle P is moving along the x-axis. At time t seconds $(t \geqslant 0)$ the velocity of P is v m/s where

$$v = t^2 - 10t + 28$$

(a) Find the velocity of P when $t = 1$

(1)

Given that the distance of P from the origin is 24 m when $t = 3$

(b) find the distance of P from the origin when $t = 5$

(5)

(c) Find the acceleration of P when $t = 9$

(2)

(d) (i) Show that there are no values of t for which P is instantaneously at rest.

(ii) Find the least magnitude of the velocity of P

(3)

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8. Question 7 · part b · 5 marks · Working and answer

Working and answer

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